

TinyCar — Command Reference

macOS · mpremote · esptool · MicroPython v1.28 (car) · v1.24.1 (remote)

1 · USB Port Detection

Run `ls /dev/cu.*` before and after plugging in to spot which port appeared.

Purpose	Command	Notes
List all serial ports	<code>ls /dev/cu.*</code>	Run before + after plugging in
Kill process holding port	<code>lsof grep usbserial</code>	Get PID from 2nd column
Force-kill stuck port	<code>kill -9 <PID></code>	Then unplug + replug ESP32
Identify chip on port	<code>python3 -m esptool --port /dev/cu.usbserial-0001 chip_id</code>	Shows MAC, chip, features

2 · esptool — Flash MicroPython

Always erase first, then flash. Port name may differ — confirm with `ls /dev/cu.*` first.

Purpose	Command	Notes
Erase entire flash	<code>python3 -m esptool --port /dev/cu.usbserial-0001 erase_flash</code>	Do before reflash

Purpose	Command	Notes
Download MicroPython v1.28 (car)	<code>curl -L https://micropython.org/resources/firmware/ESP32_GENERIC-20260406-v1.28.0.bin -o micropython.bin</code>	Car firmware version
Download MicroPython v1.24.1 (remote)	<code>curl -L https://micropython.org/resources/firmware/ESP32_GENERIC-20241129-v1.24.1.bin -o micropython.bin</code>	Remote firmware version
Flash MicroPython	<code>python3 -m esptool --port /dev/cu.usbserial-0001 --baud 460800 write_flash -z 0x1000 micropython.bin</code>	~28s, confirms hash

3 · mpremote — File Operations

Use `fs cp` to upload files. The colon prefix (`:`) means destination on the board.

Purpose	Command	Notes
List files on board	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 fs ls</code>	Shows all files + sizes
Upload file as main.py	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 fs cp main.py :main.py</code>	Runs on boot
Upload + reset (one step)	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 fs cp main.py :main.py + reset</code>	Upload then reboot
Upload supporting module	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 fs cp car_ble_receiver.py :car_ble_receiver.py</code>	Upload without rename
Delete a file	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 exec "import os; os.remove('main.py')"</code>	Remove file from board
Reset board	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 reset</code>	Hard reset via mpremote

4 · mpremote — REPL

The REPL is the live Python prompt on the board. Ctrl+C stops any running script.

Purpose	Command	Notes
Open REPL	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 repl</code>	Live console
Exit REPL	<code>Ctrl+]</code> or <code>Ctrl+X</code>	Returns to Terminal
Stop running script	<code>Ctrl+C</code>	Interrupt — drops to >>>
Get BLE MAC address	<code>import bluetooth; b=bluetooth.BLE(); b.active(True); print(b.config('mac'))</code>	Paste at >>> prompt
Check I2C device	<code>from machine import SoftI2C, Pin; i=SoftI2C(sda=Pin(21), scl=Pin(22)); print(i.scan())</code>	[41] = 0x29 = TCS34725

5 · Two-Terminal Workflow (Car + Remote simultaneously)

Open two Terminal windows — one per board. Both boards must be plugged in at the same time.

Purpose	Command / Value	Notes
Terminal 1 — Car	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 repl</code>	Watch car output
Terminal 2 — Remote	<code>python3 -m mpremote connect /dev/cu.usbserial-6 repl</code>	Watch remote output

Purpose	Command / Value	Notes
Car port (typical)	/dev/cu.usbserial-0001	MAC: 30:76:f5:e7:69: 04
Remote port (typical)	/dev/cu.usbserial-6	MAC: 5a:36:94:df:94: 8c (BLE)

6 · BLE Quick Checks

Run these at the REPL prompt to verify BLE is working before flashing full firmware.

Purpose	Command	Notes
Read car BLE MAC	<pre>import bluetooth; b=bluetooth.BLE(); b.active(True); print(b.config('mac'))</pre>	Car → 5a:36:94:df:94: 8c
Start car advertising	<pre>from car_ble_receiver import BLERemote; b = BLERemote()</pre>	Paste at car >>>
Remote scans + connects	Automatic on boot — check console for 'Connected to TinyCar!'	Within ~5 seconds
Car MAC in remote firmware	<pre>CAR_MAC = "5a:36:94:df:94:8c" # in remote_main.py</pre>	Already set correctly

7 · Full Flash Sequence — Car ESP32 (port: /dev/cu.usbserial-0001)

Use this sequence when starting fresh or after a crash. Uploads both car_ble_receiver.py and main.py.

Step	Command	Notes
Step 1 — Erase	<code>python3 -m esptool --port /dev/cu.usbserial-0001 erase_flash</code>	
Step 2 — Flash MP	<code>python3 -m esptool --port /dev/cu.usbserial-0001 --baud 460800 write_flash -z 0x1000 micropython.bin</code>	v1.28 for car
Step 3 — Upload BLE module	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 fs cp car_ble_receiver.py :car_ble_receiver.py</code>	
Step 4 — Upload main	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 fs cp main_29052026_1252.py :main.py + reset</code>	Use latest timestamped file
Step 5 — Watch output	<code>python3 -m mpremote connect /dev/cu.usbserial-0001 repl</code>	Should see 'Ready!'

8 · Full Flash Sequence — Remote ESP32 (port: /dev/cu.usbserial-6)

Remote runs v1.24.1 (not v1.28). Make sure car is already advertising before opening remote REPL.

Step	Command	Notes
Step 1 — Erase	<code>python3 -m esptool --port /dev/cu.usbserial-6 erase_flash</code>	
Step 2 — Flash MP	<code>python3 -m esptool --port /dev/cu.usbserial-6 --baud 460800 write_flash -z 0x1000 micropython.bin</code>	v1.24.1 for remote
Step 3 — Upload main	<code>python3 -m mpremote connect /dev/cu.usbserial-6 fs cp remote_main.py :main.py + reset</code>	
Step 4 — Watch output	<code>python3 -m mpremote connect /dev/cu.usbserial-6 repl</code>	Should scan + connect

⚠ Port names (/dev/cu.usbserial-0001, -6) can change — always confirm with `ls /dev/cu.*` after plugging in. ⚠ VS Code / MicroPico holds the port — close it or use MicroPico: Disconnect before running `mpremote` commands.